



Stochastic Processes

Lecture 17

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Hitting Times

Definition (Hitting Time)

Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}} \stackrel{\text{MC}}{\sim} (\mu, P)$ be DTMC on a finite state space $\mathcal{X}$. Fix $y \in \mathcal{X}$.

Let $\tau_y^{(0)} := 1$, and let

$$\tau_y^{(k)} = \inf\{n > \tau_y^{(k-1)} : X_n = y\}, \quad k \in \mathbb{N}.$$

The random variable $\tau_y^{(k)}$ is called the **k th hitting time of state y** .

Note: At each hitting time $\tau_y^{(k)}$, we have

$$X_{\tau_y^{(k)}} = y.$$

Lemma (Hitting Time is Adapted)

For each $k \in \mathbb{N}$, the random variable $\tau_y^{(k)}$ is adapted to the process $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$, i.e.,

$$\{\tau_y^{(k)} = m\} \in \sigma(X_1, \dots, X_m) \quad \forall m \in \mathbb{N}.$$

Hitting Times and Recurrence Times

- By definition, we have

$$\tau_Y^{(1)} < \tau_Y^{(2)} < \tau_Y^{(3)} < \dots$$

- For each k , although $\tau_Y^{(k)}$ is adapted to the natural filtration of $\mathbf{X}$, it may not be a stopping time
- If $\mathbb{P}(\{\tau_Y^{(k)} < +\infty\}) = 1$, then $\tau_Y^{(k)}$ is a stopping time adapted to the natural filtration of $\mathbf{X}$

Recurrence Time

For each $k \in \mathbb{N}$, the increment

$$R_Y^{(k)} := \tau_Y^{(k+1)} - \tau_Y^{(k)}$$

is called the **k th recurrence time**.

Hitting Times and Recurrence Times

- If $\tau_y^{(k)}$ is a stopping time, then by the strong Markov property, we have

$$X_{\tau_y^{(k)}+n} \perp\!\!\!\perp \left(\tau_y^{(k)}, X_1, \dots, X_{\tau_y^{(k)}} \right) \mid \left(\tau_y^{(k)}, X_{\tau_y^{(k)}} \right) \quad \forall n \in \mathbb{N}.$$

- Because $X_{\tau_y^{(k)}} = y$ deterministically, it follows that if $\tau_y^{(k)}$ is a stopping time, then

$$X_{\tau_y^{(k)}+n} \perp\!\!\!\perp \left(\tau_y^{(k)}, X_1, \dots, X_{\tau_y^{(k)}} \right) \mid \tau_y^{(k)} \quad \forall n \in \mathbb{N}.$$

- Observe that

$$\{R_y^{(k)} = m\} = \{\tau_y^{(k+1)} - \tau_y^{(k)} = m\} = \{X_{\tau_y^{(k)}+1} \neq y, \dots, X_{\tau_y^{(k)}+m-1} \neq y, X_{\tau_y^{(k)}+m} = y\}$$

Lemma (Independence of Recurrence Time in Relation to its Past)

If $\tau_y^{(k)}$ is a stopping time, then

$$R_y^{(k)} \perp\!\!\!\perp \left(\tau_y^{(k)}, X_1, \dots, X_{\tau_y^{(k)}} \right) \mid \tau_y^{(k)}$$

Hitting Times and Recurrence Times

- For each $k \in \mathbb{N}$, consider the process $\mathbf{Y}^{(k)} = \{Y_n^{(k)}\}_{n \in \mathbb{N}}$ defined as

$$Y_n^{(k)} := X_{\tau_Y^{(k)} + n}, \quad n \in \mathbb{N}.$$

- If $\tau_Y^{(k)}$ is a stopping time, then for all $n \in \mathbb{N}$ and $(y_1, \dots, y_n) \in \mathcal{X}^n$,

$$\begin{aligned} & \mathbb{P}\left(\{Y_1^{(k)} = y_1, \dots, Y_n^{(k)} = y_n\}\right) \\ &= \mathbb{P}\left(\{X_{\tau_Y^{(k)} + 1} = y_1, \dots, X_{\tau_Y^{(k)} + n} = y_n\}\right) \\ &= \mathbb{P}\left(\{X_{\tau_Y^{(k)} + 1} = y_1, \dots, X_{\tau_Y^{(k)} + n} = y_n, \tau_Y^{(k)} < +\infty\}\right) \\ &= \sum_m \mathbb{P}\left(\{X_{\tau_Y^{(k)} + 1} = y_1, \dots, X_{\tau_Y^{(k)} + n} = y_n, \tau_Y^{(k)} = m\}\right) = \mathbb{P}\left(\{X_2 = y_1, \dots, X_{n+1} = y_n\} \mid \{X_1 = y\}\right). \end{aligned}$$

Lemma (Renewal of Markov Chain at Hitting Times)

If $\tau_Y^{(k)}$ is a stopping time, then the FDD of $\mathbf{Y}^{(k)}$ is identical to that of $\mathbf{X} \stackrel{\text{MC}}{\sim} (\mathbf{1}_{\{y\}}, P)$.

Hitting Times and Recurrence Times

- If $\tau_Y^{(k)}$ is a stopping time for each k , we can deduce from Lemma 2 that $\{R_Y^{(k)}\}_{k \in \mathbb{N}}$ is a sequence of **independent** random variables
- If $\tau_Y^{(k)}$ is a stopping time for each k , we can deduce from Lemma 3 that $\{R_Y^{(k)}\}_{k \in \mathbb{N}}$ is a sequence of **identically distributed** random variables
- If $\tau_Y^{(2)}$ is a stopping time, then $\tau_Y^{(k)}$ is a stopping time for each k

Proof:

- If $\tau_Y^{(2)}$ is a stopping time, then $\tau_Y^{(1)}$ is a stopping time, and hence $R_Y^{(1)} = \tau_Y^{(2)} - \tau_Y^{(1)}$ is finite a.s.
- If $\tau_Y^{(2)}$ is a stopping time, then as a consequence of Lemma 3, we have

$$\tau_Y^{(3)} - \tau_Y^{(2)} = R_Y^{(2)} \stackrel{\text{d.}}{=} R_Y^{(1)} = \tau_Y^{(3)} - \tau_Y^{(2)}.$$

Hence, it follows that $R_Y^{(2)}$ is finite a.s.

- The above arguments may be extended via induction for all k

On Recurrence Times

- For $x, y \in \mathcal{X}$, let

$$f_{xy}^{(n)} := \mathbb{P}\left(\{\tau_y^{(1)} = n\} \mid \{X_1 = x\}\right), \quad n \in \{2, 3, \dots\}.$$

Interpretation of $f_{xy}^{(n)}$: **probability of first hit to state y at time n when starting in state x**

- For $x, y \in \mathcal{X}$, let

$$f_{xy} = \mathbb{P}\left(\{\tau_y^{(1)} < +\infty\} \mid \{X_1 = x\}\right) = \sum_n f_{xy}^{(n)}.$$

- Interpretation of f_{xy} : **probability of ever hitting state y conditioned on $\{X_1 = x\}$**
- Interpretation of $(1 - f_{xy})$: **probability of never hitting state y conditioned on $\{X_1 = x\}$**

- Observe that

$$\mathbb{P}\left(\{\tau_y^{(1)} < +\infty\}\right) = \sum_{x \in \mathcal{X}} \mathbb{P}\left(\{\tau_y^{(1)} < +\infty\} \mid \{X_1 = x\}\right) \cdot \mathbb{P}\left(\{X_1 = x\}\right) = \sum_{x \in \mathcal{X}} f_{xy} \cdot \mathbb{P}\left(\{X_1 = x\}\right).$$

- If $f_{xy} = 1$ for all $x \in \mathcal{X}$, then

$$\mathbb{P}(\{\tau_y^{(1)} < +\infty\}) = 1,$$

and hence $\tau_y^{(1)}$ is a stopping time

Recurrent and Transient States

Definition (Recurrent and Transient States)

A state $x \in \mathcal{X}$ is called **recurrent** if $f_{xx} = 1$.

A state $x \in \mathcal{X}$ is called **transient** if $f_{xx} < 1$.

Remarks:

- The vector

$$\left[f_{xx}^{(2)}, f_{xx}^{(3)}, \dots, 1 - f_{xx} \right]$$

defines a valid PMF on $\{2, 3, \dots\} \cup \{+\infty\}$

- The above PMF is called **first recurrence time distribution**
- One can then ask: what is the mean under the above PMF?

Mean Recurrence Time

Definition (Mean Recurrence Time)

The **mean recurrence time** of a state $x \in \mathcal{X}$ is denoted by μ_{xx} and is defined by

$$\mu_{xx} := \mathbb{E} \left[\tau_x^{(1)} \mid \{X_1 = x\} \right].$$

Remarks:

- If x is transient, then $\mu_{xx} = +\infty$
- If x is recurrent, then

$$\mu_{xx} = \sum_{n \geq 2} n f_{xx}^{(n)}.$$

- If $\mu_{xx} < +\infty$, then x is called a **positive recurrent** state
- If $\mu_{xx} = +\infty$, then x is called a **null recurrent** state
- A state x can be only one of 3 kinds: transient, positive recurrent, or null recurrent

Visitation Probabilities

Theorem (Visitation Probabilities)

For any two states $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) = \begin{cases} , & k = 0, \\ , & k \in \mathbb{N}. \end{cases}$$

Visitation Probabilities

Theorem (Visitation Probabilities)

For any two states $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) = \begin{cases} 1 - f_{xy}, & k = 0, \\ f_{xy} f_{yy}^{k-1} (1 - f_{yy}), & k \in \mathbb{N}. \end{cases}$$

Proof of Theorem 5

For any $k \in \mathbb{N}$, we have

$$\begin{aligned}
 & \mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) \\
 &= \mathbb{P}\left(\{\tau_y^{(1)} < +\infty, \tau_y^{(2)} < +\infty, \dots, \tau_y^{(k)} < +\infty, \tau_y^{(k+1)} = +\infty\} \mid \{X_1 = x\}\right) \\
 &= \mathbb{P}\left(\{\tau_y^{(1)} < +\infty, \tau_y^{(2)} - \tau_y^{(1)} < +\infty, \dots, \tau_y^{(k)} - \tau_1^{(k-1)} < +\infty, \tau_y^{(k+1)} - \tau_1^{(k)} = +\infty\} \mid \{X_1 = x\}\right) \\
 &= f_{xy} \times \left(\prod_{\ell=2}^k f_{y\gamma}\right) \times (1 - f_{y\gamma}).
 \end{aligned}$$

Corollary 1

Theorem (Visitation Probabilities)

For any two states $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) = \begin{cases} 1 - f_{xy}, & k = 0, \\ f_{xy} f_{yy}^{k-1} (1 - f_{yy}), & k \in \mathbb{N}. \end{cases}$$

Corollary 1

For any $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited **finitely many times**}\} \mid \{X_1 = x\}\right) = \begin{cases} 1, & f_{yy} < 1, \\ 1 - f_{xy}, & f_{yy} = 1. \end{cases}$$

Corollary 2

Theorem (Visitation Probabilities)

For any two states $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) = \begin{cases} 1 - f_{xy}, & k = 0, \\ f_{xy} f_{yy}^{k-1} (1 - f_{yy}), & k \in \mathbb{N}. \end{cases}$$

Corollary 2

For any $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited infinitely many times}\} \mid \{X_1 = x\}\right) = \begin{cases} 0, & f_{yy} < 1, \\ f_{xy}, & f_{yy} = 1. \end{cases}$$

Corollary 3

Theorem (Visitation Probabilities)

For any two states $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) = \begin{cases} 1 - f_{xy}, & k = 0, \\ f_{xy} f_{yy}^{k-1} (1 - f_{yy}), & k \in \mathbb{N}. \end{cases}$$

Corollary 3

For any $x, y \in \mathcal{X}$,

$$\mathbb{E}\left[\underbrace{\#\text{visits to state } y}_{N_y} \mid \{X_1 = x\}\right] = \begin{cases} \frac{f_{xy}}{1-f_{yy}}, & f_{yy} < 1, \\ +\infty, & f_{yy} = 1, f_{xy} > 0. \end{cases}$$

Transience and Recurrence in a Nutshell

Theorem (Complete Characterisation of Transience and Recurrence)

Let $\mathbf{X} \stackrel{\text{MC}}{\sim} (\mu, P)$ be a homogeneous DTMC on a finite state space $\mathcal{X}$ with TPM P .

1. A state $y \in \mathcal{X}$ is transient if and only if: $\mathbb{E}[N_y \mid \{X_1 = y\}] < +\infty$.
2. A state $y \in \mathcal{X}$ is recurrent if and only if: $\mathbb{E}[N_y \mid \{X_1 = y\}] = +\infty$.

Remarks:

- Note that $N_y = \sum_{n \geq 2} \mathbf{1}_{\{X_n = y\}}$, and hence

$$\mathbb{E}[N_y \mid \{X_1 = y\}] = \mathbb{E}\left[\sum_{n \geq 2} \mathbf{1}_{\{X_n = y\}} \mid \{X_1 = y\}\right] \stackrel{\text{MCT}}{=} \sum_{n \geq 2} \mathbb{P}(\{X_n = y\} \mid \{X_1 = y\}).$$

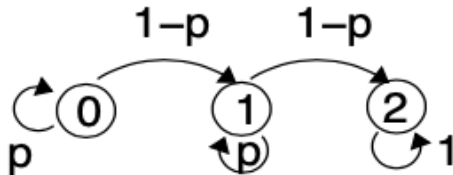
- If $\mathbb{E}[N_y \mid \{X_1 = y\}] < +\infty$, then by Borel-Cantelli lemma, y is visited only finitely many times a.s. when starting in y

Transience and Recurrence in a Nutshell

- A transient state y is visited only finitely many times a.s., no matter the initial state
- If y is recurrent and the Markov chain starts in y , then y will be visited infinitely many times a.s.
- If y is recurrent and the Markov chain starts in some initial state $x \neq y$ for which $f_{xy} = 1$, then y will be visited infinitely many times a.s.
- If $|\mathcal{X}| < +\infty$, then all states in $\mathcal{X}$ cannot be transient

Example

- Consider a time-homogeneous DTMC with following transition graph.



Compute f_{00} , f_{11} , and f_{22} .

When $p \in (0, 1)$, classify the states into transient, positive recurrent, and null recurrent.